

COMBINATIONAL LOGIC DESIGN

Module.2

CDA 3103C

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Mesut Ozdag, Ph.D.

K-map minimization (con't)

- ◆ Obtain the complement by covering 0s with subcubes

AB		A			
		00	01	11	10
C	0	1 0	0 2	0 6	1 4
	1	0 1	0 3	1 7	1 5
		B			

$$F(A,B,C) = \sum m(0,4,5,7) \\ = B'C' + AC$$

$$F'(A,B,C) = \sum m(1,2,3,6) \\ = A'C + BC'$$

AB		A			
		00	01	11	10
C	0	1	0	1	1
	1	0	0	1	1

$$F(A,B,C) = ???$$

$$F'(A,B,C) = ???$$

$$F = \sum m(0,4,5,6,7) = A + B'C'$$

$$F' = \sum m(1,2,3) = A'C + A'B = A'(B+C)$$

Definition of terms for two-level simplification

- Implicant
 - Single element of ON-set or DC-set or any group of these elements that can be combined to form a subcube
- Prime implicant
 - Implicant that can't be combined with another to form a larger subcube
- Essential prime implicant
 - Prime implicant is essential if it alone covers an element of ON-set
 - Will participate in ALL possible covers of the ON-set
 - DC-set used to form prime implicants but not to make implicant essential
- Objective:
 - Grow implicant into prime implicants (minimize literals per term)
 - Cover the ON-set with as few prime implicants as possible (minimize number of product terms)

Examples to illustrate terms

		A		
	0	X	1	0
	1	1	1	0
	1	0	1	1
C	0	0	1	1
		B		

6 prime implicants:

$A'B'D$, BC' , AC , $A'C'D$, AB , $B'CD$

essential

minimum cover: $AC + BC' + A'B'D$

5 prime implicants:

BD , ABC' , ACD , $A'BC$, $A'C'D$

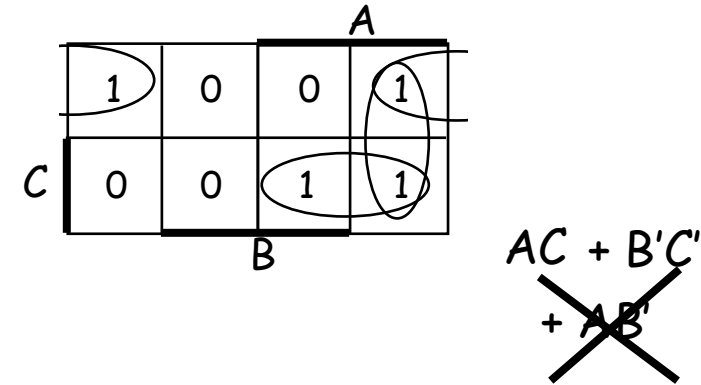
essential

minimum cover: 4 essential implicants

		A		
	0	0	1	0
	1	1	1	0
C	0	1	1	1
	0	1	0	0
		B		

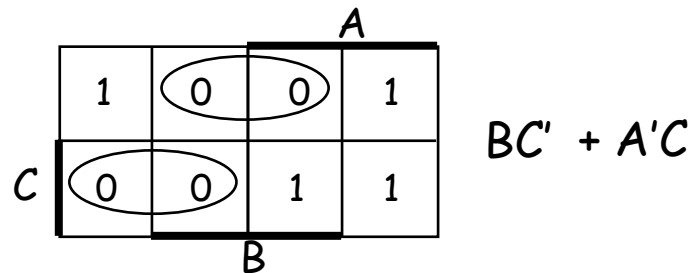
Karnaugh map examples

- $f(A,B,C) = \Sigma m(0,4,5,7)$



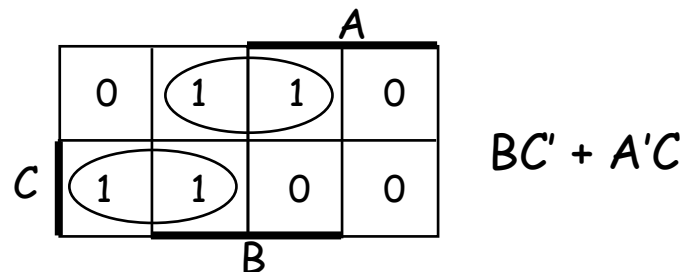
- Can we also determine f' ?

- Option 1:



We can obtain the complement of the function by covering the 0s instead of 1s

- Option 2:



We can obtain the complement by replacing 1's with 0's and vice versa

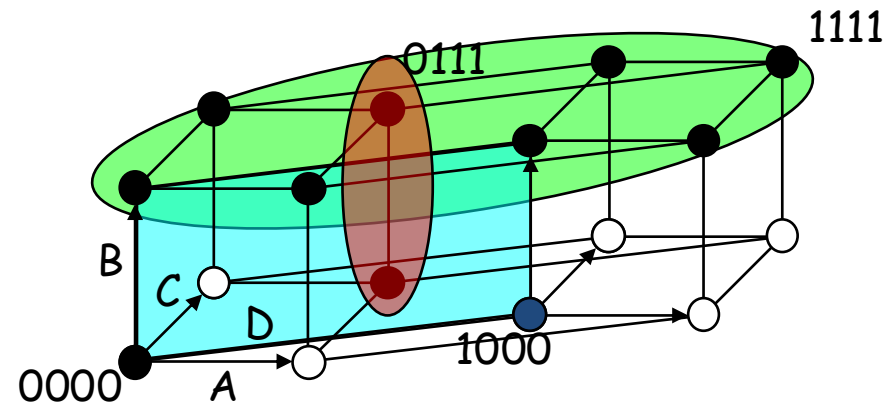
Karnaugh map: 4-variable example

- $F(A,B,C,D) = \Sigma m(0,2,3,5,6,7,8,10,11,14,15)$

F =

$$C + A'BD + B'D'$$

		A			
		1	0	0	1
		0	1	0	0
	C	1	1	1	1
		1	1	1	1
	B				



find the smallest number of the largest possible
subcubes to cover the ON-set
(fewer terms with fewer inputs per term)

Karnaugh maps: don't cares

- $f(A,B,C,D) = \Sigma m(1,3,5,7,9) + d(6,12,13)$
 - without don't cares
 - $f = A'D + B'C'D$

				A	
		0	0	X	0
		1	1	X	1
		1	1	0	0
		0	X	0	0
		B			
C	D				

Karnaugh maps: don't cares

- $f(A,B,C,D) = \Sigma m(1,3,5,7,9) + d(6,12,13)$
 - $f = A'D + B'C'D$ without don't cares
 - $f = A'D + C'D$ with don't cares

A			
0	0	X	0
1	1	X	1
1	1	0	0
0	X	0	0
B			
C			
D			

by using don't care as a "1",
a 2-cube can be formed
rather than a 1-cube to cover
this node

don't cares can be treated as
1s or 0s
depending on which is more
advantageous

Algorithm for two-level simplification

- To get the minimum sum-of-products expression from a Karnaugh map:
 - Step 1: choose an element of the ON-set
 - Step 2: find "maximal" groupings of 1s and Xs adjacent to that element
 - consider top/bottom row, left/right column, and corner adjacencies
 - this forms prime implicants (number of elements always a power of 2)
 - Repeat Steps 1 and 2 to find all prime implicants
 - Step 3: revisit the 1s in the K-map
 - if covered by single prime implicant, it is essential, and participates in final cover
 - 1s covered by essential prime implicant do not need to be revisited
 - Step 4: if there remain 1s not covered by essential prime implicants
 - select the smallest number of prime implicants that cover the remaining 1s

Algorithm for two-level simplification (example)

				A
	X	1	0	1
	0	1	1	1
	0	X	X	0
C	0	1	0	1
				B

				A
	X	1	0	1
	0	1	1	1
	0	X	X	0
C	0	1	0	1
				B

2 primes around $A'BC'D'$

				A
	X	1	0	1
	0	1	1	1
	0	X	X	0
C	0	1	0	1
				B

2 primes around $ABC'D$

				A
	X	1	0	1
	0	1	1	1
	0	X	X	0
C	0	1	0	1
				B

3 primes around $AB'C'D'$

				A
	X	1	0	1
	0	1	1	1
	0	X	X	0
C	0	1	0	1
				B

2 essential primes

				A
	X	1	0	1
	0	1	1	1
	0	X	X	0
C	0	1	0	1
				B

minimum cover (3 primes)



MESUT OZDAG, PH.D.
MESUT.OZDAG@UCF.EDU